

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

AD1101-1 – COMPUTER ORGANIZATION

(AI&DS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

- _____ is generally used to increase the apparent size of physical memory.
 - Secondary memory
 - Hard-disk
 - Virtual memory
 - Disks
- SPEC** stands for _____.
 - Standard Performance Evaluation Code
 - System Processing Enhancing Code
 - System Performance Evaluation Corporation
 - Standard Processing Enhancement Corporation
- The decoded instruction is stored in _____.
 - IR
 - PC
 - Register
 - MDR
- The addressing mode/s, which uses the PC instead of a general purpose register is _____.
 - Indexed with offset
 - Relative
 - Direct
 - Both Indexed with offset and direct
- In the case of Zero-address instruction method, the operands are stored in _____.
 - Registers
 - Registers
 - Push down stack
 - Push down stack
- What does the end instruction do?
 - It ends the generation of a signal
 - It starts a new instruction fetch cycle and resets the counter
 - It ends the complete generation process
 - It is used to shift the control to the processor
- To read the control words sequentially _____ is used.
 - PC
 - micro PC
 - IR
 - MAR
- What is the result of the Booth encoding for a pair of adjacent bits '01' in the multiplier?
 - 0
 - 0
 - 1
 - 1
- For the addition of large integers, most of the systems make use of _____.
 - Fast adders
 - Full adders
 - Carry look-ahead adders
 - Carry save adder
- The final addition sum of the numbers, 0110 & 0110 is _____.
 - 1101
 - 1111
 - 1001
 - 1010
- The method used to reduce the maximum number of summands by half is _____.
 - Fast multiplication
 - Bit-pair recording
 - Quick multiplication
 - Restoring method
- In memory-mapped I/O _____.
 - The I/O devices and the memory share the same address space
 - The I/O devices have a separate address space
 - The memory and I/O devices have an associated address space
 - A part of the memory is specifically set aside for the I/O operation
- The interrupt servicing mechanism in which the requesting device identifies itself to the processor to be serviced is _____.
 - Polling
 - Vectored interrupts
 - Interrupt nesting
 - Simultaneous requesting

14. _____ method is used to establish priority by serially connecting all devices that request an interrupt.
 A) Vectored-interrupting B) Polling
 C) Priority D) Daisy chain
15. The technique where the controller is given complete access to main memory is _____
 A) Cycle stealing B) Memory stealing
 C) Burst mode D) Memory Con
16. The DMA transfer is initiated by _____
 A) Processor B) The process being executed
 C) I/O devices D) OS
17. Which of the following is the fastest means of memory access for CPU?
 A) Registers B) Cache
 C) Main memory D) Virtual Memory
18. In ROM, each bit is a combination of the address variables is called _____
 A) Memory unit B) Storage class
 C) Data word D) Address
19. The full form of EEPROM is _____
 A) Erasable Electrically Programmable ROMs B) Electrically Erasable Programmable ROMs
 C) Electrically Erasable Programming ROMs D) Electrically Erasable Programmed ROMs
20. In set-associative technique, the blocks are grouped into _____ sets.
 A)4 B)8
 C)12 D)6

PART - B: DESCRIPTIVE ANSWER QUESTIONS

- | Unit – I | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | | |
|---|-------------------------------|-------------------------------|--------------------------|-----|------------|-----------|---|------------|-----------|---|------------|-----------|---|------------|-----------|---|-----|---|---|
| 1. a) List and explain the steps needed to execute the machine instruction ADD LOCA, R0 in terms of transfer between the components of processor, memory & some control commands. Assume the instruction is stored in memory location 'INSTR'. Remember to update the PC content.
b) What is performance measurement? Explain the overall SPEC rating for the computer in a program suit. Assuming that the reference computer is ultra SPARCIO workstation with 300 MHz ultra SPARC processor. A company has to purchase 1000 new computers hence ordered testing of new computer with SPEC 2000. Following observation were made: <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <thead> <tr> <th>Programs</th> <th>Runtime on reference computer</th> <th>Run time on new computer</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>70 minutes</td> <td>5 minutes</td> </tr> <tr> <td>2</td> <td>60 minutes</td> <td>4 minutes</td> </tr> <tr> <td>3</td> <td>75 minutes</td> <td>5 minutes</td> </tr> <tr> <td>4</td> <td>68 minutes</td> <td>2 minutes</td> </tr> </tbody> </table> The company system manger will place the order for purchasing new computers only if the overall SPEC rating is at least 12. After the said test will the system manager place order for purchase of new computer? | Programs | Runtime on reference computer | Run time on new computer | 1 | 70 minutes | 5 minutes | 2 | 60 minutes | 4 minutes | 3 | 75 minutes | 5 minutes | 4 | 68 minutes | 2 minutes | 8 | L*2 | 1 | 1 |
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| 1 | 70 minutes | 5 minutes | | | | | | | | | | | | | | | | | |
| 2 | 60 minutes | 4 minutes | | | | | | | | | | | | | | | | | |
| 3 | 75 minutes | 5 minutes | | | | | | | | | | | | | | | | | |
| 4 | 68 minutes | 2 minutes | | | | | | | | | | | | | | | | | |
| 2. a) What is addressing mode? Develop instruction in indexed addressing mode for the following operation- Compute sum of scores of 3 tests and store the result in the location SUM1, SUM2 and SUM3 of n students.
b) Illustrate parameter passing using stack in subroutine, with example. | 8 | L3 | 1 | 1 | | | | | | | | | | | | | | | |
| 3. a) With a figure, explain the single bus organization of the datapath inside a processor.
b) Explain hardwired control and micro programmed control unit. | 8 | L2 | 2 | 1 | | | | | | | | | | | | | | | |

Unit – II

- | | | | | | |
|-------|---|---|----|---|---|
| 4. a) | Explain booth algorithm and apply booth algorithm to multiply signed numbers -13 and 9. | 8 | L3 | 3 | 1 |
| b) | Define Bus arbitration. Explain in detail any one approach of bus arbitration | 8 | L2 | 4 | 1 |
| 5. a) | Develop a 4 bit carry look ahead adder and explain in detail. | 8 | L3 | 3 | 1 |
| b) | Illustrate DMA controller with a neat diagram. | 8 | L2 | 4 | 1 |
| 6. a) | Draw circuit arrangement for binary division. Perform 1000 / 11 using non restoring method. | 8 | L3 | 3 | 1 |
| b) | With suitable diagram explain the following with respect to interrupts: | | | | |
| | i) Interrupt Nesting ii) Simultaneous Request | 8 | L2 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|---|
| 7. a) | Draw and explain internal organization of 2M X 8 asynchronous DRAM chip. | 8 | L2 | 5 | 1 |
| b) | Explain different types of Read Only Memory. | 8 | L2 | 5 | 1 |
| 8. a) | Briefly explain any two mapping function in cache memory | 8 | L2 | 5 | 1 |
| b) | Explain the virtual memory organization. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
